Introduction into Additive Manufacturing

MAE 1001 Guest Lecture

Friday October 16th 2026

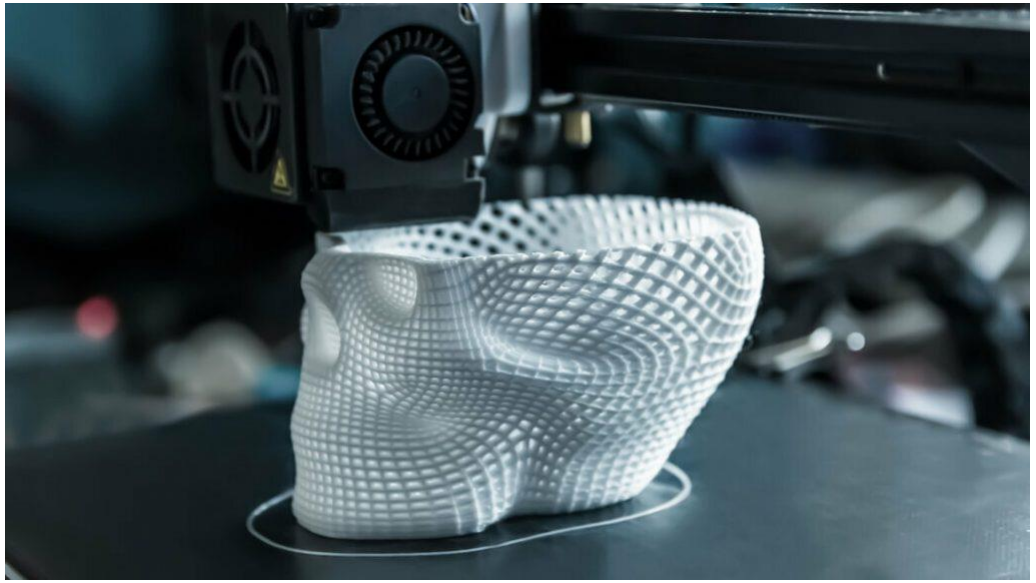
Sumner Gubisch

What is Additive Manufacturing?

Core Definition: Additive Manufacturing, often called **3D Printing**, is a revolutionary process of creating a **three-dimensional object** by **adding material layer by layer.**

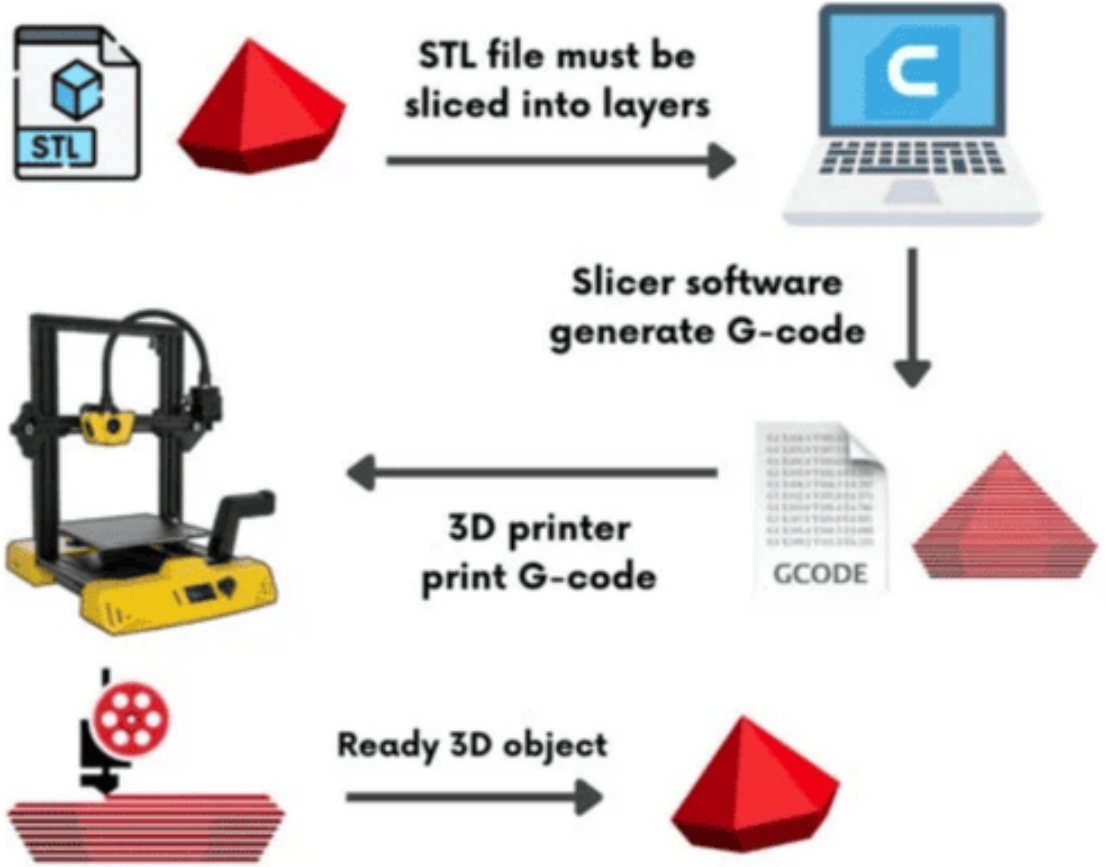


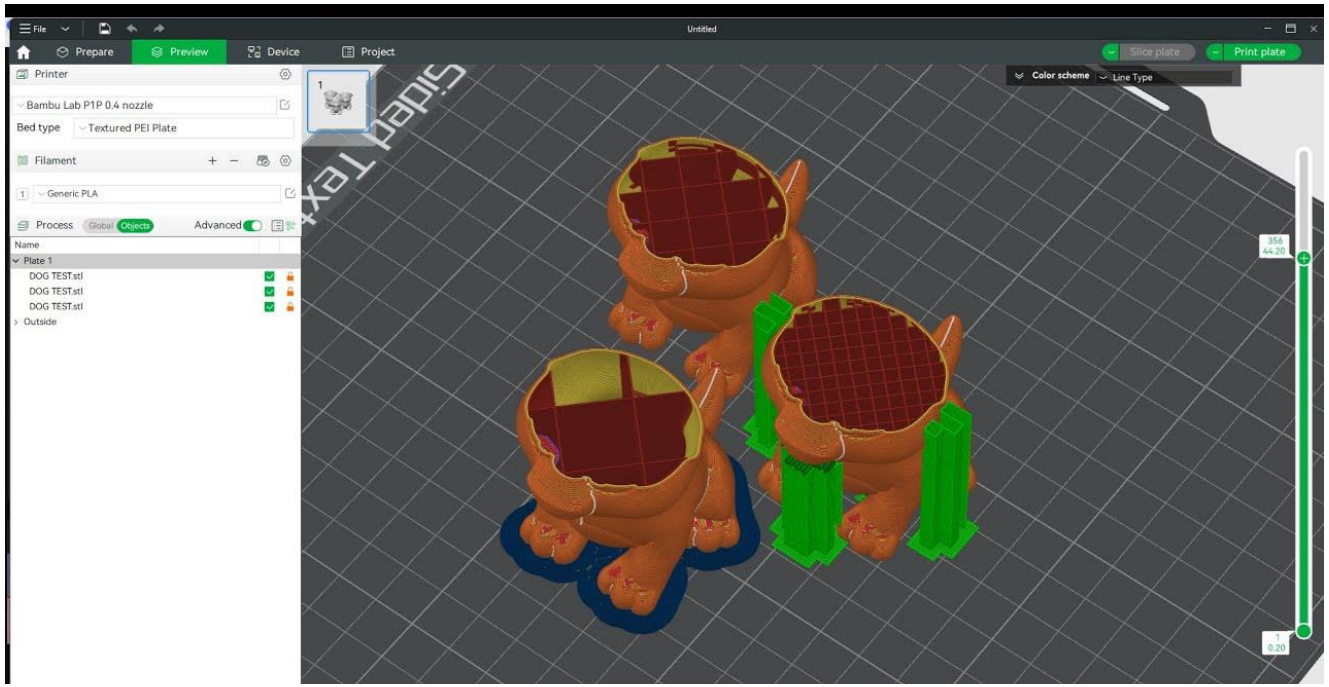
Unlike **subtractive manufacturing** (like machining), which removes material from a solid block, AM builds the object from the *bottom up*, resulting in **less waste**.

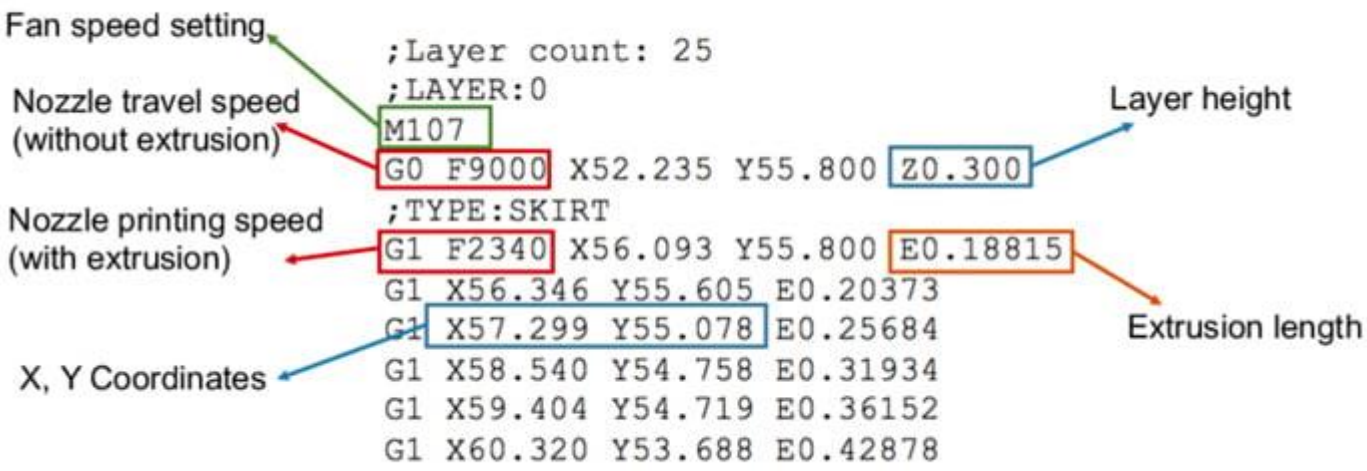


Key Benefit: This layer-by-layer approach allows for the creation of **complex geometries** and **intricate internal structures** that are impossible or too costly to produce with traditional methods.

Additive Manufacturing Slicer and G Code







Types of Additive Manufacturing

Extrusion Based Manufacturing (EBM)

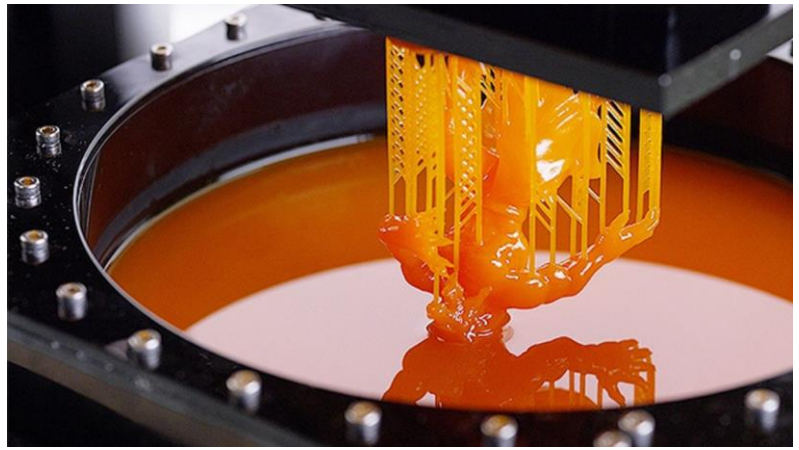


Fused Filament Fabrication (FFF)

Direct Ink Writing (DIW)



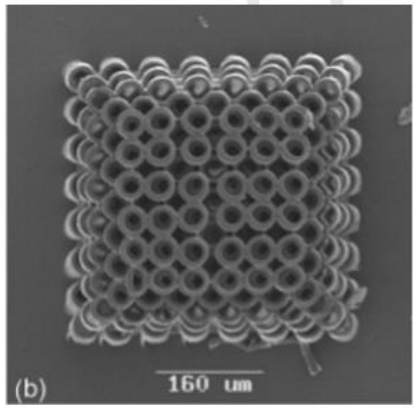
Vat Photopolymerization (VPP)



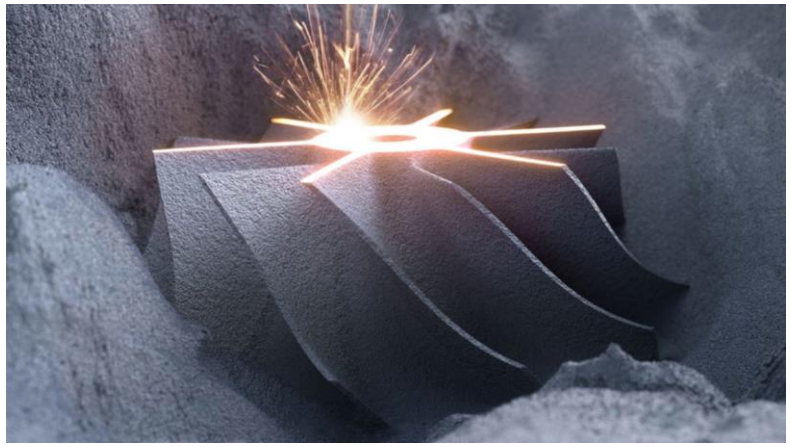
Stereolithography Apparatus (SLA)

Direct Laser Writing (DLW)





Power Based



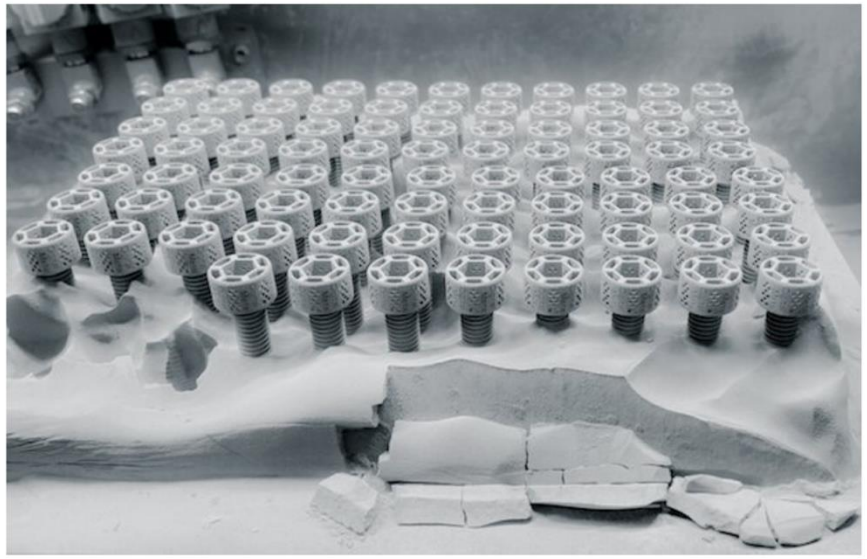
Laser Powder Bed Fusion (LPBF)

Direct Energy Deposition (DED)



Other Types of Additive Manufacturing

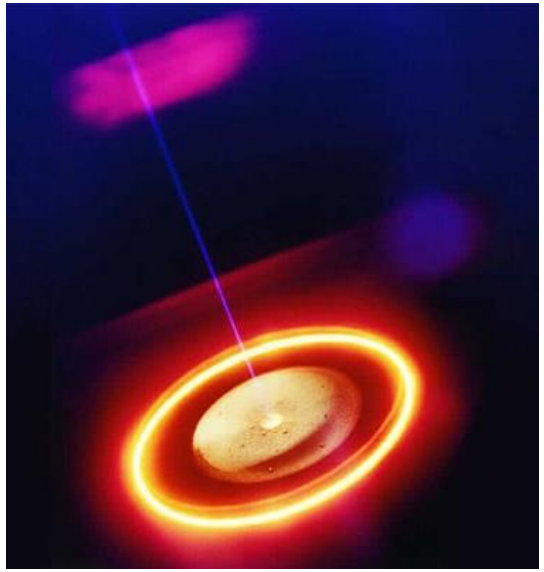
Binder Jet Printing (BJP)



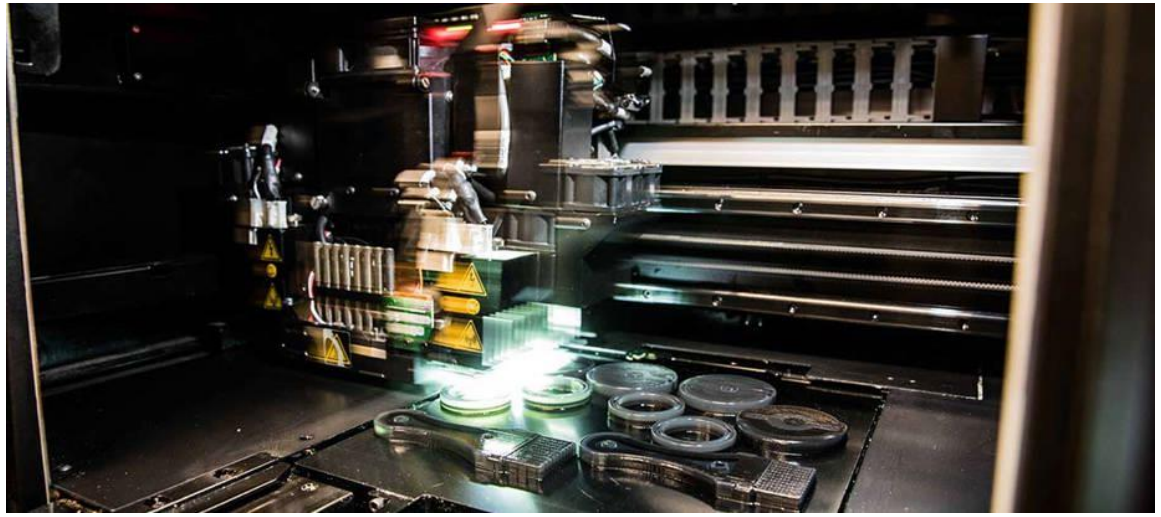
Selective Laser Sintering (SLS)



Electron Beam Melting (EBM)



Poly Jet / Ink Jet printing (IJP)



Electron Beam Additive Manufacturing (EBAM)



Advantages vs Disadvantages of Additive Manufacturing

Advantages

- Can print complex designs
- Consolidation of multiple parts
- Mass customizability
- Single Tooling cost
- Reduced material waste
  - Only put material where it is needed
- Locally producing parts
- Rapid prototyping

Disadvantages

- Anisotropy for many parts being produced.
- Limited part size
- Reduced surface finished
- Cost per item made doesn't change much
- Slow production speed
- High material cost and equipment cost
- Post processing cost

When to Use Additive Manufacturing

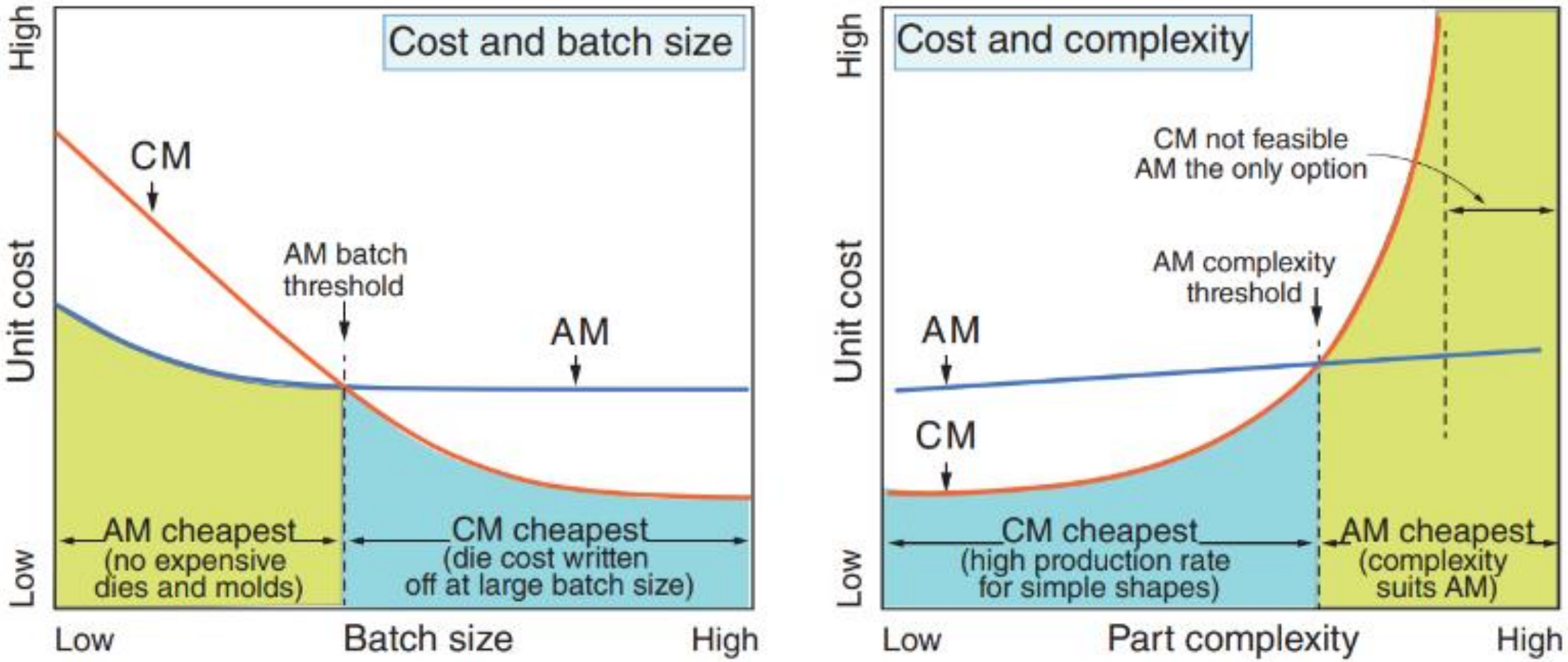
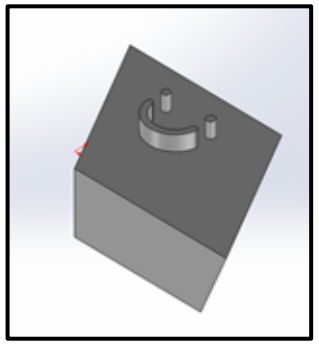
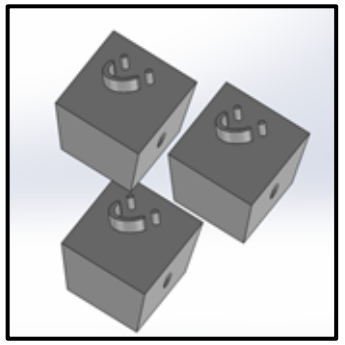


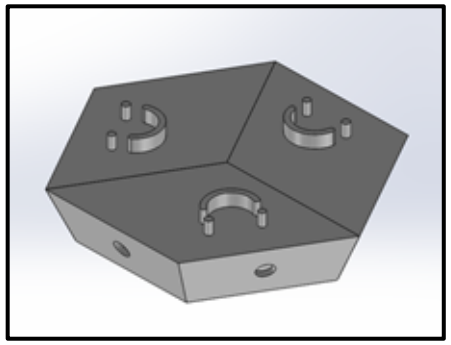
FIGURE 8.6

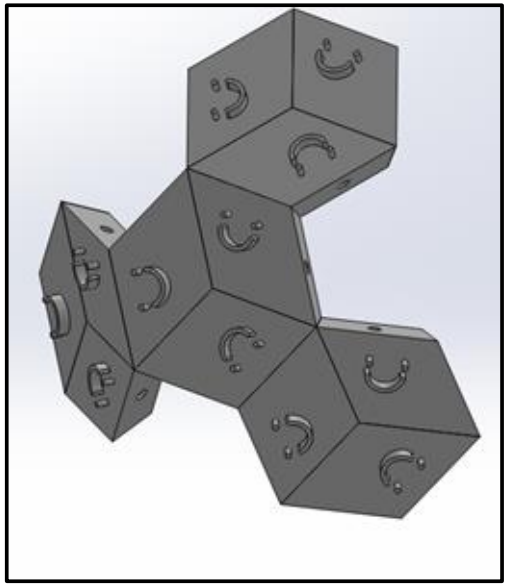
Batch size and part complexity determine the relative costs of AM and conventionally manufactured (CM) parts. *(After Leary, M, 2019. Design for Additive Manufacturing, Elsevier.).*

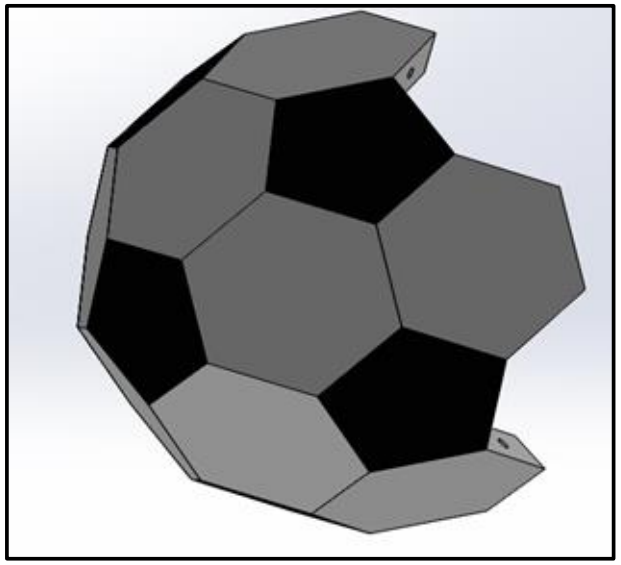
Soccer Ball Project – Tile Prints

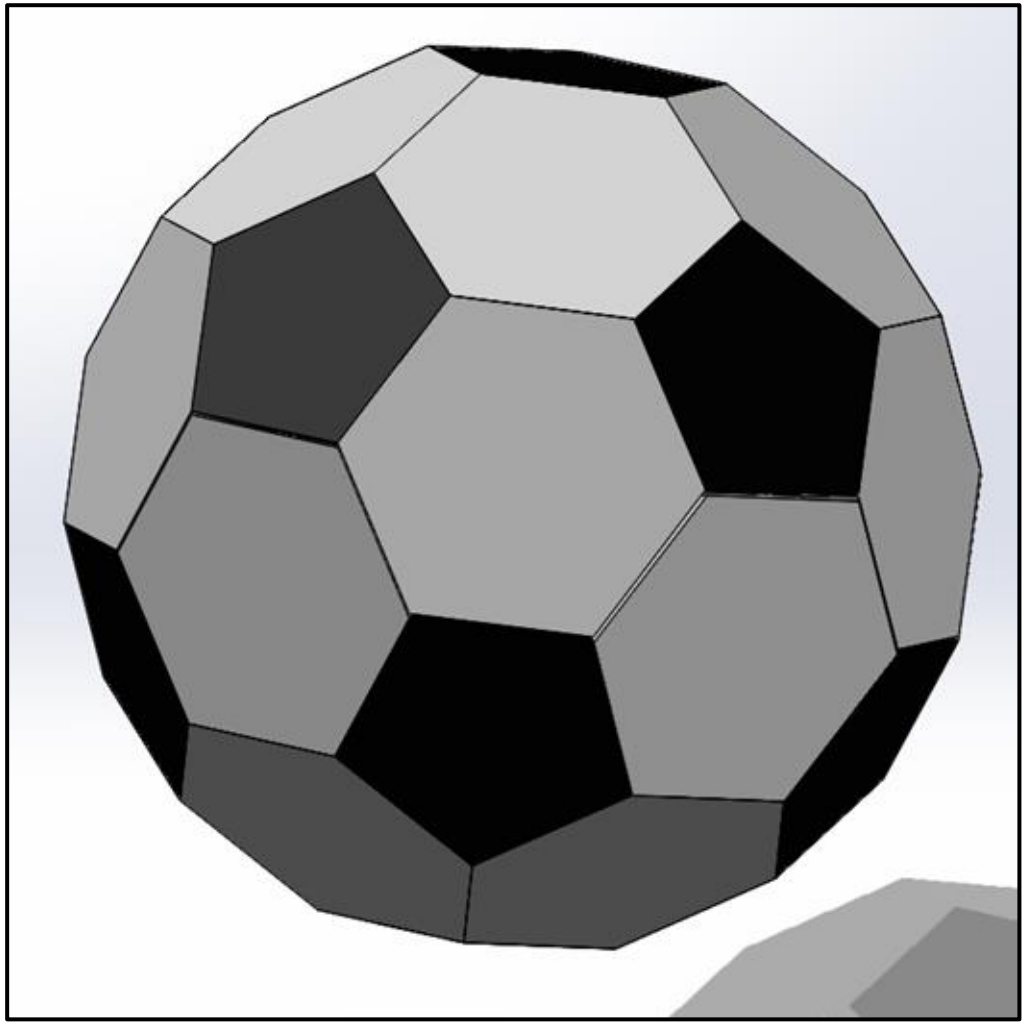








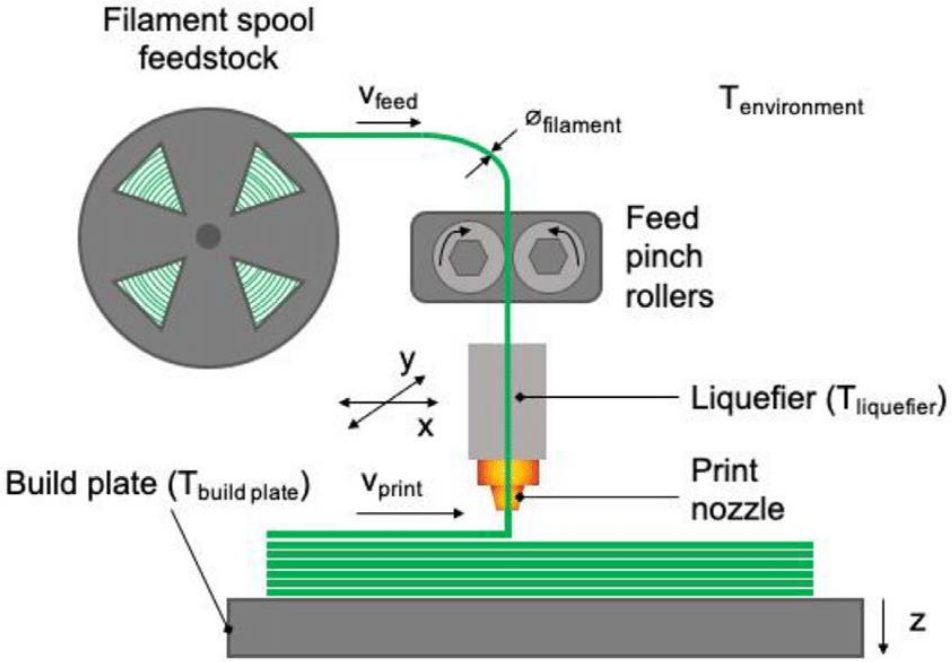


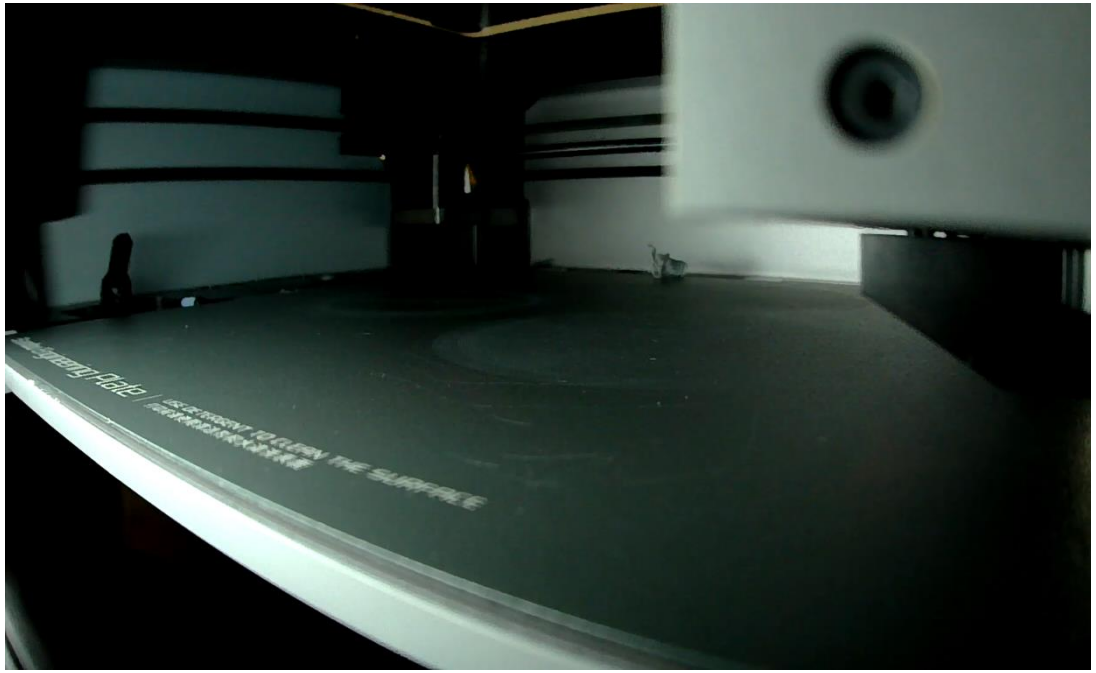


Fused Filament Fabrication (FFF)

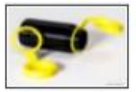
Processing Steps

1. Thermoplastic is fed through a heated nozzle
2. The melted thermoplastic is pushed through the nozzle with the unmelted filament
3. The nozzle moves in x and y directions and bed moves in z
4. Filament solidifies and a solid part is produced
5. Supports are removed as needed





Fused Filament Materials

	 ABS	 Flexible	 PLA	 HIPS	 PETG	 Nylon	 Carbon Fiber Filled	 ASA	 Polycarbonate	 Polypropylene	 Metal Filled	 Wood Filled	 PVA
	Learn More	Learn More	Learn More	Learn More	Learn More	Learn More	Learn More	Learn More	Learn More	Learn More	Learn More	Learn More	Learn More
Compare Selected	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Ultimate Strength	40 MPa	26 - 43 MPa	65 MPa	32 MPa	53 MPa	40 - 85 MPa	45 - 48 MPa	55 MPa	72 MPa	32 MPa	20 - 30 MPa	46 MPa	78 MPa
Stiffness	5 / 10	1 / 10	7.5 / 10	10 / 10	5 / 10	5 / 10	10 / 10	5 / 10	6 / 10	4 / 10	10 / 10	8 / 10	3 / 10
Durability	8 / 10	9 / 10	4 / 10	7 / 10	8 / 10	10 / 10	3 / 10	10 / 10	10 / 10	9 / 10	4 / 10	3 / 10	7 / 10
Maximum Service Temperature	98 °C	60 - 74 °C	52 °C	100 °C	73 °C	80 - 95 °C	52 °C	95 °C	121 °C	100 °C	52 °C	52 °C	75 °C
Coefficient of Thermal Expansion	90 µm/m-°C	157 µm/m-°C	68 µm/m-°C	80 µm/m-°C	60 µm/m-°C	95 µm/m-°C	57.5 µm/m-°C	98 µm/m-°C	69 µm/m-°C	150 µm/m-°C	33.75 µm/m-°C	30.5 µm/m-°C	85 µm/m-°C
Density	1.04 g/cm³	1.19 - 1.23 g/cm³	1.24 g/cm³	1.03 - 1.04 g/cm³	1.23 g/cm³	1.06 - 1.14 g/cm³	1.3 g/cm³	1.07 g/cm³	1.2 g/cm³	0.9 g/cm³	2 - 4 g/cm³	1.15 - 1.25 g/cm³	1.23 g/cm³
Price (per kg)	$10 - $40	$30 - $70	$10 - $40	$24 - $32	$20 - $60	$25 - $65	$30 - $80	$38 - $40	$40 - $75	$60 - $120	$50 - $120	$25 - $55	$40 - $110
Printability	8 / 10	6 / 10	9 / 10	6 / 10	9 / 10	8 / 10	8 / 10	7 / 10	6 / 10	4 / 10	7 / 10	8 / 10	5 / 10
Extruder Temperature	220 - 250 °C	225 - 245 °C	190 - 220 °C	230 - 245 °C	230 - 250 °C	220 - 270 °C	200 - 230 °C	235 - 255 °C	260 - 310 °C	220 - 250 °C	190 - 220 °C	190 - 220 °C	185 - 200 °C
Bed temperature	95 - 110 °C	45 - 60 °C	45 - 60 °C	100 - 115 °C	75 - 90 °C	70 - 90 °C	45 - 60 °C	90 - 110 °C	80 - 120 °C	85 - 100 °C	45 - 60 °C	45 - 60 °C	45 - 60 °C
Heated Bed	Required	Optional	Optional	Required	Required	Required	Optional	Required	Required	Required	Optional	Optional	Required
Recommended Build Surfaces	Kapton Tape, ABS Slurry	PEI, Painter's Tape	Painter's Tape, Glue Stick, Glass Plate, PEI	Glass Plate, Glue Stick, Kapton Tape	Glue Stick, Painter's Tape	Glue Stick, PEI	Painter's Tape, Glue Stick, Glass Plate, PEI	Glue Stick, PEI	PEI, Commercial Adhesive, Glue Stick	Packing Tape, Polypropylene Sheet	Painter's Tape, Glue Stick, PEI	Painter's Tape, Glue Stick, PEI	PEI, Painter's Tape
Other Hardware Requirements	Heated Bed, Enclosure Recommended	Part Cooling Fan	Part Cooling Fan	Heated Bed, Enclosure Recommended	Heated Bed, Part Cooling Fan	Heated Bed, Enclosure Recommended, May Require All Metal Hotend	Part Cooling Fan	Heated Bed	Heated Bed, Enclosure Recommended, All Metal Hotend	Heated Bed, Enclosure Recommended, Part Cooling Fan	Wear Resistant or Stainless Steel Nozzle, Part Cooling Fan	Part Cooling Fan	Heated Bed, Part Cooling Fan

Fused Filament Fabrication Post Processing

Standard Support



Standard Support

Tree Support

Dissolvable Support



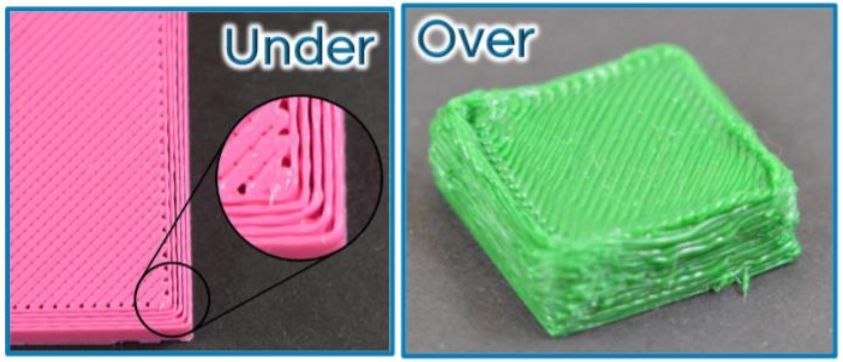
PVA / HIPS

Can dissolve in water or IPA

FFF Challenges

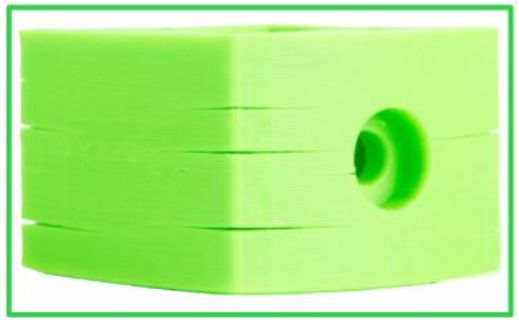
Under or Over Extrusion

- Extruder wells not optimized
- Clogged



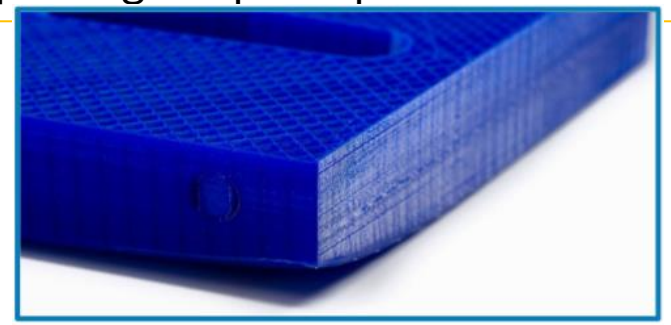
Delamination

- Caused by layers shrinking causing cracking
- Material dependent



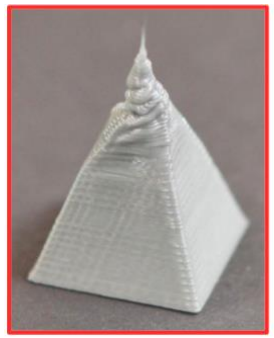
Warping

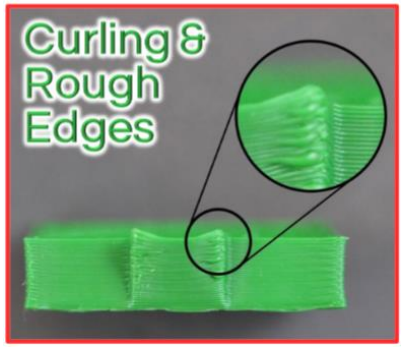
- Caused by the build plate being to cold and the stress from the cooling peeling the part up



Overheating

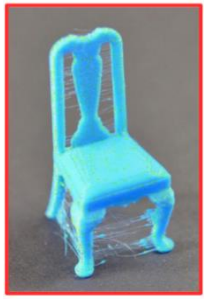
- Material doesn't cool down enough before next layer and it starts to melt

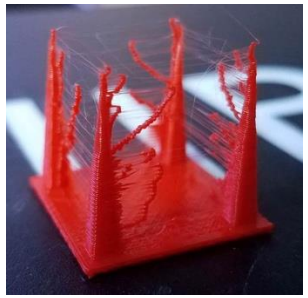




Retraction

- Filament is too hot or extruder movement is moving to fast
- Results in stringing





Elephant Footing

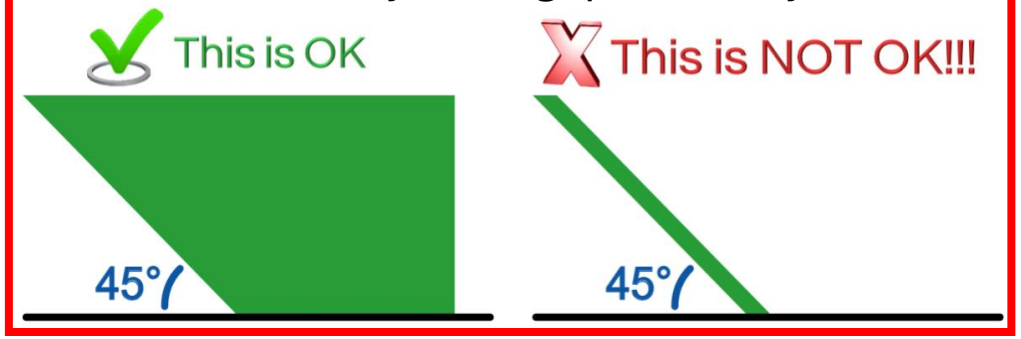
- Caused by the build plate being to hot and melting the base layers



FFF Printing Tips

Support Considerations

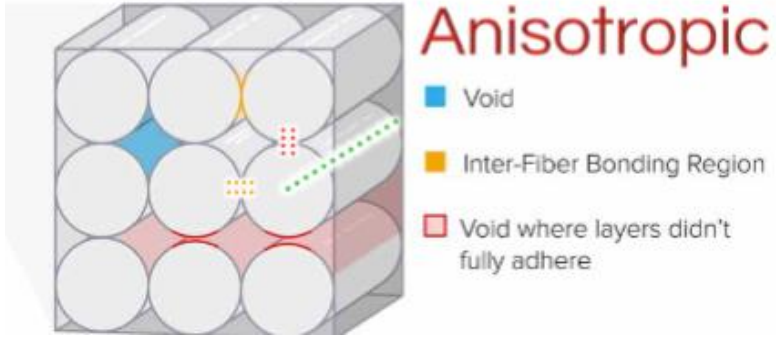
- > 45° support recommended
- More than 45° with extra material
- Manually designed support .4m gap horizontally 1mm gap vertically



Tolerance

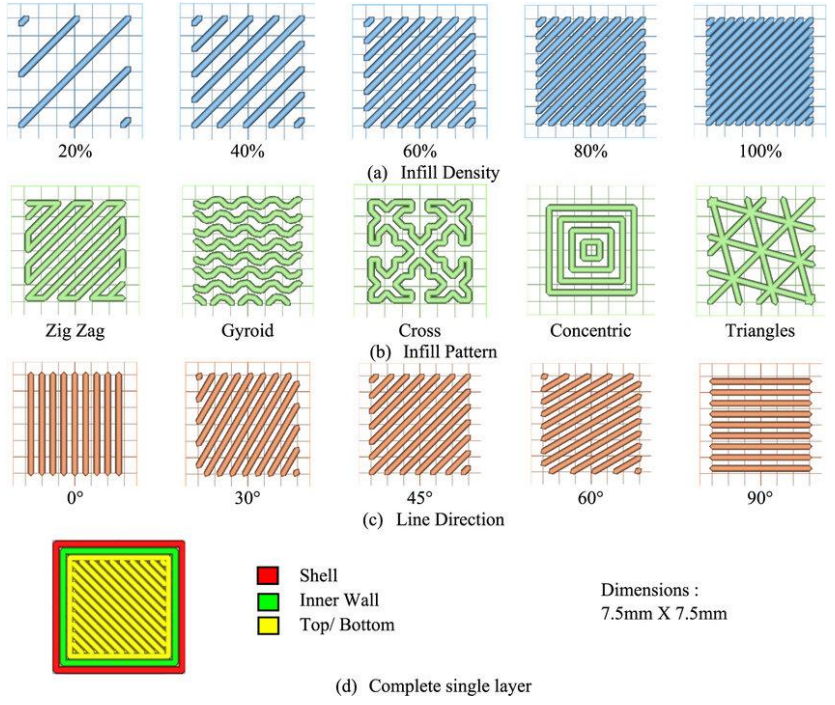
- 0.1 mm vertically
- 0.2 mm horizontally
- 0.5 mm for engraved
- 0.9 mm for embossed
- 1.8 mm pins
- 2mm for holes





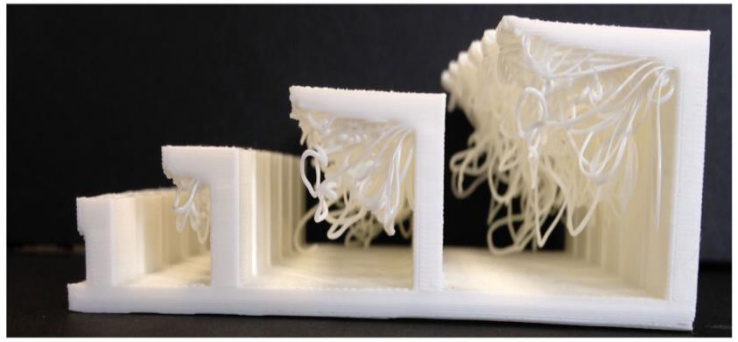
Infill

- Infill important for strength
- Number of shells matter more



Overhangs

- Sensitive to overhangs
- > 1cm results in stringing



Bridging

- Can be a challenge >10 cm
- As it cools it can contract and pull the bridge upwards



Questions?